Thadomal Shahani Engineering College

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EXPERIMENT 6: To study and Implement Agile Software Project lifecycle using Jira with DevOps Pipeline Integration.

Aim: To implement Agile Software Project lifecycle using Jira with DevOps Pipeline Integration.

Theory:

1. Fundamentals of Agile and Scrum Methodology

Agile is a software development approach that focuses on iterative development, continuous collaboration, customer feedback, and delivering working software in smaller increments. Scrum is an Agile framework that divides the project into fixed-duration cycles called Sprints, usually lasting 1 to 4 weeks.

A Scrum team consists of three main roles:

- Product Owner: Sets product objectives, prioritizes requirements, and maintains the Product Backlog.
- Scrum Master: Conducts daily stand-up meetings, resolves obstacles, and ensures that Scrum practices are followed.
- Development Team: A group of members with different skills who work together to deliver usable software increments during each sprint.

2. Jira Work Breakdown Structure (WBS) and Hierarchy

Jira Software helps organize project requirements into a structured hierarchy, making it easier to plan, assign, and track work from larger goals to smaller tasks.

- Roadmap / Timeline: Displays the overall project schedule, milestones, and dependencies over several months.
- Epics: Represent large features or initiatives that may require multiple sprints to complete, such as improving customer service.
- User Stories: Describe specific features from the user's perspective, such as providing a live chat facility.

- Tasks and Sub-tasks: Break user stories into smaller technical activities that developers can complete.
- Issue Dependencies: Show relationships between tasks, such as one task being blocked until another is completed.

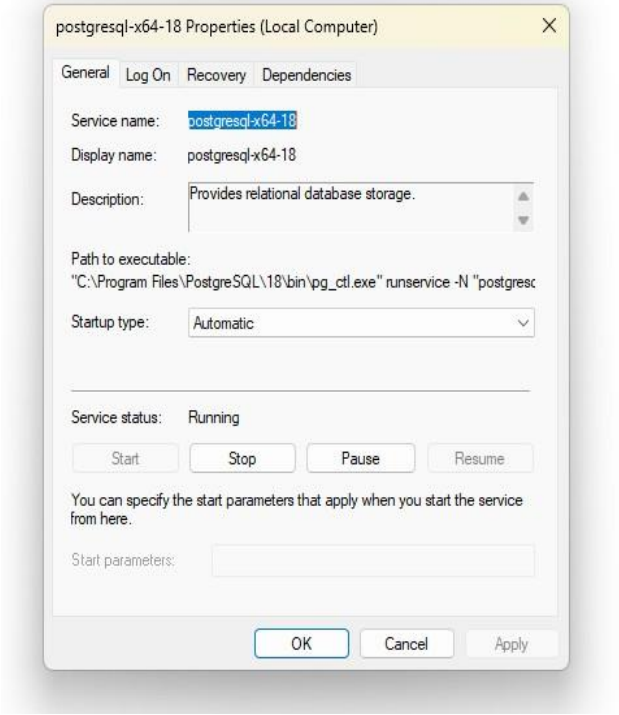
3. Sprint Execution and Board Customization

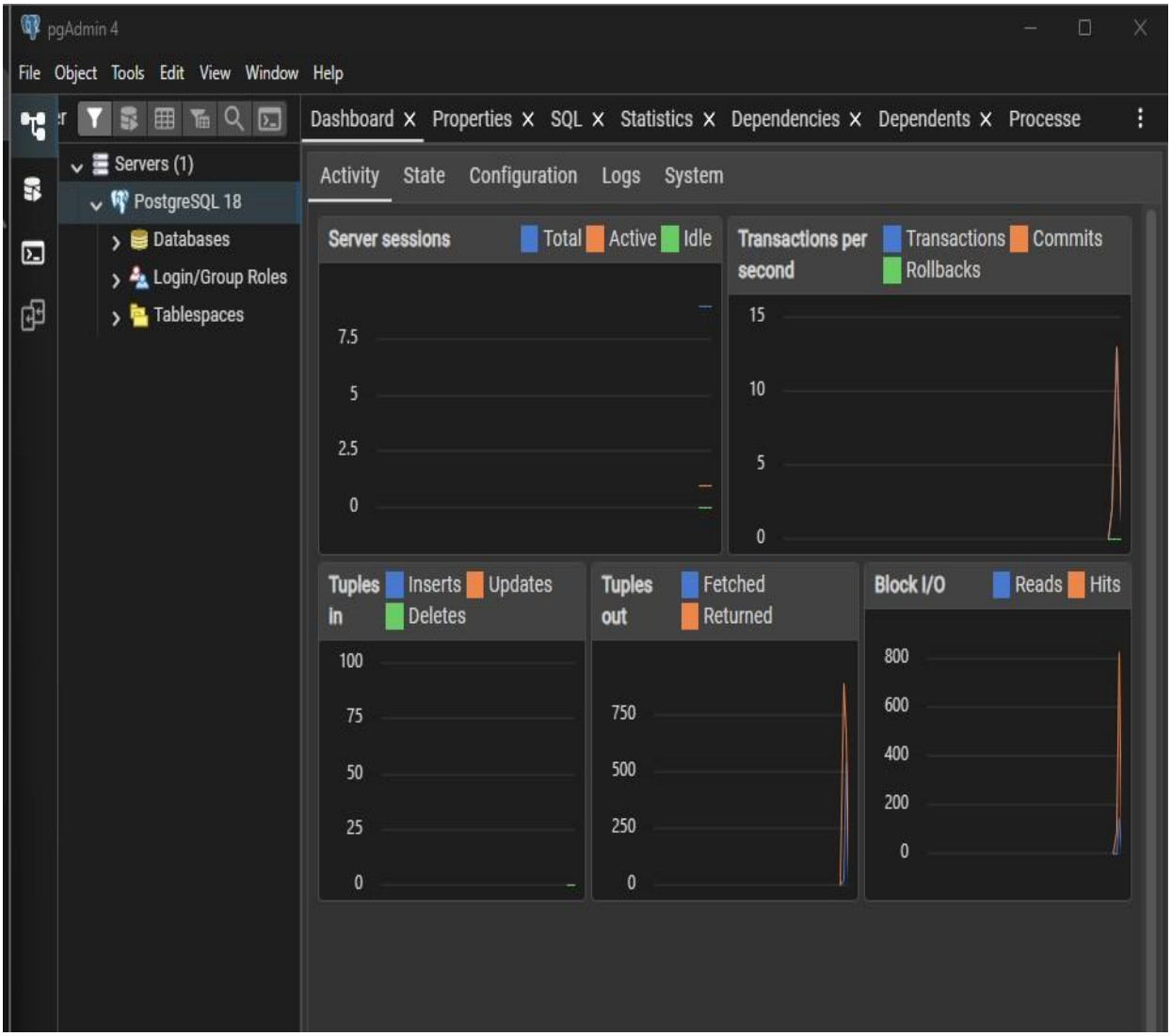
During sprint planning, selected issues are moved from the Product Backlog into the active Sprint. The Scrum board helps the team monitor progress through different stages of development using status columns and swimlanes.

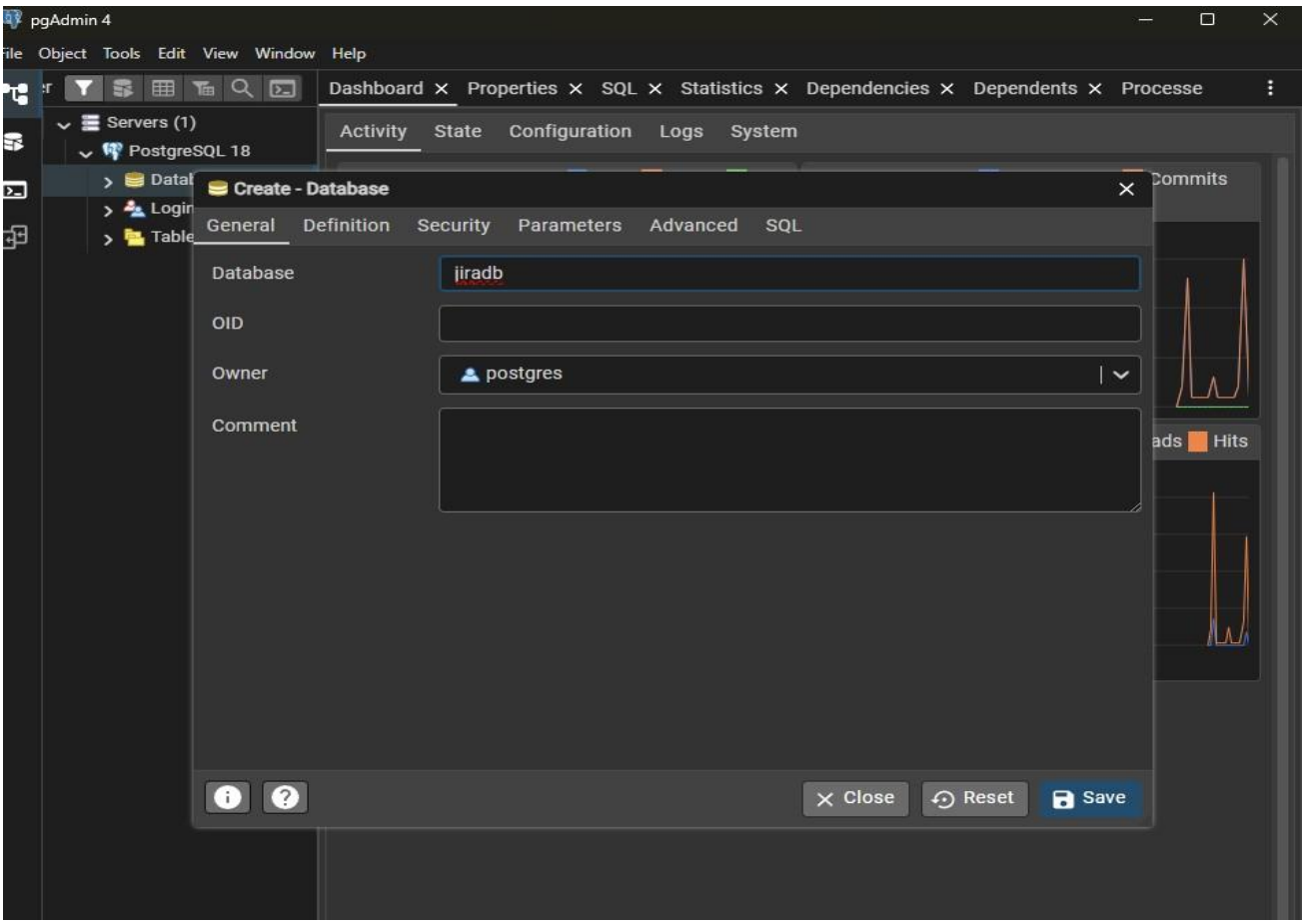
- Standard Columns: Commonly include To Do, In Progress, and Done to represent the current status of each task.
- Custom Workflow Columns: Additional stages, such as Review and Testing, can be added to ensure code reviews and quality checks are completed before a task is marked as Done.

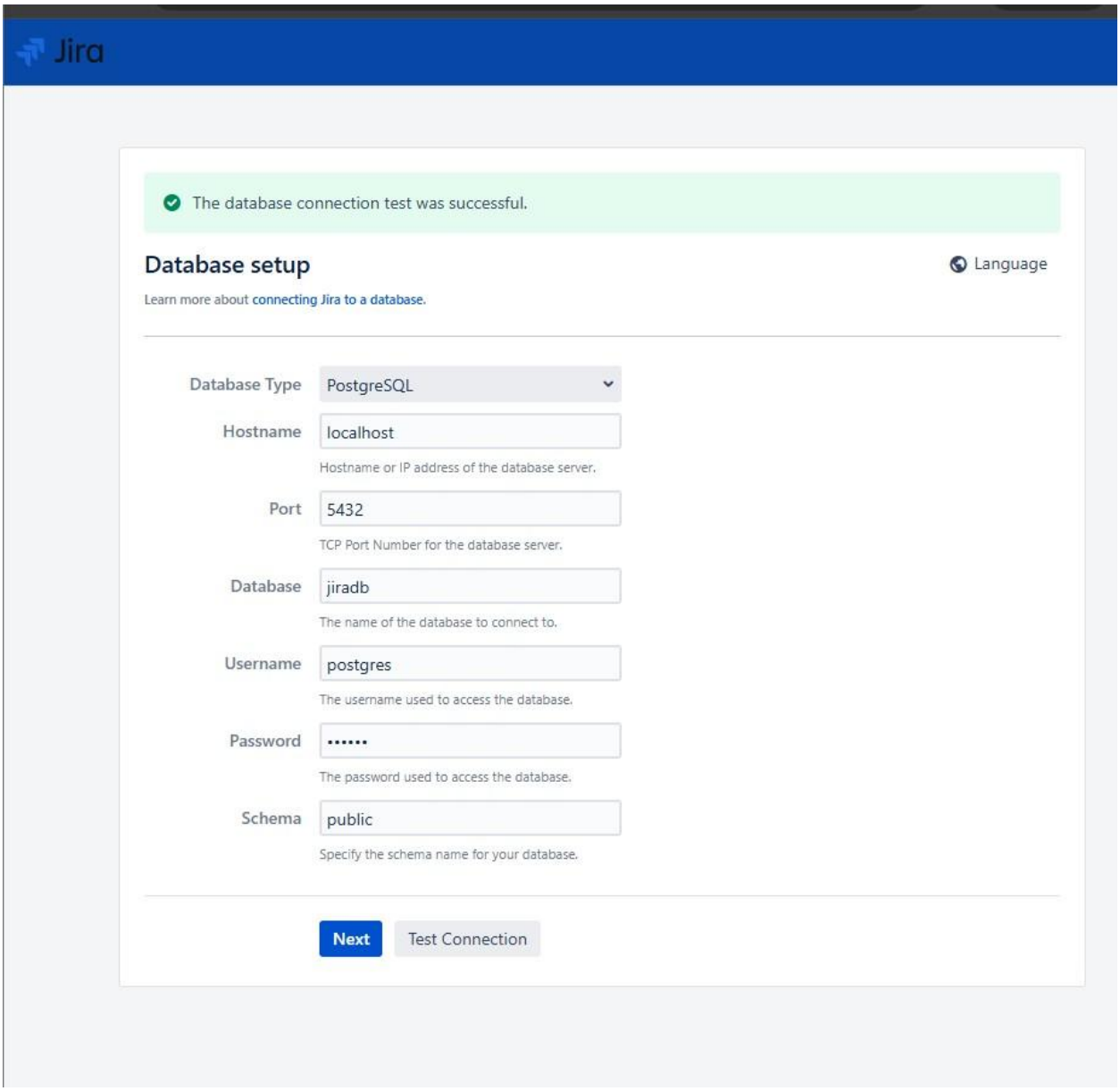
These boards improve transparency, help identify delays, and allow the team to track sprint progress efficiently.

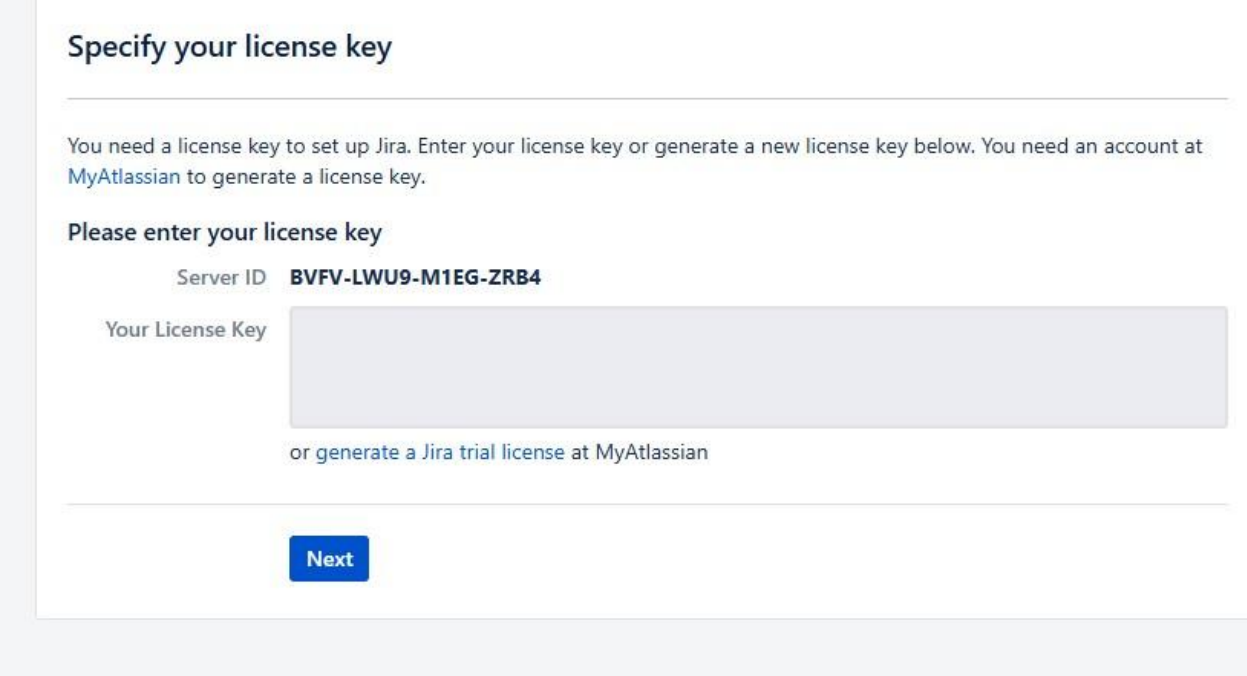
Implementation:




pgAdmin 4
File Object Tools Edit View Window Help
Dashboard X Properties X SQL X Statistics X Dependencies X Dependents X Processe
Servers (1)
PostgreSQL 18
Databases
Login/Group Roles
Tablespaces
Activity State Configuration Logs System
Server sessions Total Active Idle
7.5
5
2.5
0
Transactions per second Transactions Commits Rollbacks
15
10
5
0
Tuples in Inserts Updates Deletes
100
75
50
25
0
Tuples out Fetched Returned
750
500
250
0
Block I/O Reads Hits
800
600
400
200
0


pgAdmin 4
File Object Tools Edit View Window Help
Dashboard
Properties
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Statistics
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Dependents
Processe
Activity
State
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Logs
System
Servers (1)
PostgreSQL 18
Commits
Hits
Create - Database
General
Definition
Security
Parameters
Advanced
SQL
Database
jiradb
OID
Owner
postgres
Comment
Close
Reset
Save


Jira
The database connection test was successful.
Database setup
Language
Learn more about connecting Jira to a database.
Database Type
PostgreSQL
Hostname
localhost
Hostname or IP address of the database server.
Port
5432
TCP Port Number for the database server.
Database
jiradb
The name of the database to connect to.
Username
postgres
The username used to access the database.
Password
••••••
The password used to access the database.
Schema
public
Specify the schema name for your database.
Next
Test Connection



Conclusion:

In this experiment, an Agile Scrum project was created and managed using Jira Software. A software development project named **Sheify Website Development** was set up with a structured hierarchy of Epics, User Stories, and related Tasks. Dependencies between issues were added to represent practical project requirements. Sprint planning was also carried out by defining sprint goals, moving backlog items to an active Scrum board, and customizing the workflow by adding a Review stage. Overall, the experiment provided practical experience in backlog management, sprint execution, progress tracking, and handling task dependencies using Jira.